Grad-B and Curvature Drift Benchmark in a Simple Analytical Magnetic Mirror

1. Purpose of the benchmark

This benchmark extends the simple analytical magnetic-mirror orbit problem to test whether the particle pusher reproduces the expected guiding-center drifts. In a nonuniform magnetic field, a charged particle does not simply gyrate around a fixed field line. Its gyrocenter also drifts across the magnetic field. In an axisymmetric mirror, the two most important collisionless drifts are the grad-B drift, caused by spatial variation in the magnetic-field strength, and the curvature drift, caused by curvature of the magnetic-field direction.

The goal of this test is to compare the measured azimuthal gyrocenter drift from a full Boris-pushed particle orbit against standard guiding-center drift theory. The magnetic field is analytical, axisymmetric, and generated from a uniform guide field plus two end solenoids. This makes it a clean code-verification problem.

2. Magnetic-field model

The magnetic field is built from a uniform guide field and two finite end solenoids centered at

$$z = \pm 2.0 \text{ m} \quad .$$

The field parameters were chosen as

$$B_{\text{guide}} = 0.30 \text{ T} \quad ,$$

$$B_{\text{mirror}} \approx 1.50 \text{ T} \quad .$$

The code uses Cartesian field components,

$$B_x, B_y, B_z,$$

rather than cylindrical components inside the particle pusher. The axisymmetric field is first defined through an analytical on-axis field $B_0(z)$. The off-axis field is then approximated using the lowest-order divergence-free expansion,

$$B_r(r,z) \simeq -\frac{r}{2}\frac{dB_0}{dz},$$

$$B_z(r,z) \simeq B_0(z) - \frac{r^2}{4}\frac{d^2B_0}{dz^2}.$$

The Cartesian components used by the pusher are then

$$B_x = B_r\frac{x}{r},$$

$$B_y = B_r\frac{y}{r},$$

$$B_z = B_z.$$

This gives a simple three-dimensional magnetic field that still contains field-strength gradients and field-line curvature.

3. Particle and numerical setup

The test particle is a 14 keV deuteron launched off axis at

$$x_0 = 0.220 \text{ m} \quad , y_0 = 0, z_0 = 0.$$

The pitch angle is

$$\alpha_0 = 60°.$$

An off-axis launch is important because the grad-B and curvature drifts in an axisymmetric mirror are primarily azimuthal. If the particle were launched exactly on axis, the azimuthal guiding-center drift would not be as clearly represented as a circular precession.

The particle is advanced using the Boris pusher with no electric field. The Boris pusher preserves the speed well in static magnetic fields and is therefore a good choice for isolating magnetic drift physics.

The launch field strength is

$$B_{\text{launch}} \approx 0.3273 \text{ T} \qquad .$$

The corresponding launch Larmor radius is

$$\rho_L \approx 0.0640 \text{ m} \qquad .$$

4. Guiding-center drift theory

The total first-order magnetic drift velocity is the sum of the grad-B drift and curvature drift,

$$\mathbf{v}_d = \mathbf{v}_{\nabla B} + \mathbf{v}_c.$$

The grad-B drift is

$$\mathbf{v}_{\nabla B} = \frac{mv_\perp^2}{2qB^3}\mathbf{B} \times \nabla B.$$

This drift occurs because the particle experiences a slightly different magnetic field strength across one gyro-orbit. The result is a net drift perpendicular to both the magnetic field and the magnetic-field gradient.

The curvature drift is

$$\mathbf{v}_c = \frac{mv_\parallel^2}{qB}\mathbf{b} \times \boldsymbol{\kappa},$$

where

$$\mathbf{b} = \frac{\mathbf{B}}{B}$$

is the magnetic-field unit vector and

$$\boldsymbol{\kappa} = (\mathbf{b} \cdot \nabla)\mathbf{b}$$

is the magnetic-field curvature vector. This drift appears because a particle moving along a curved magnetic field line experiences a centrifugal effect.

In this axisymmetric mirror, both drifts are mainly in the azimuthal direction. Therefore, the most useful scalar comparison is the azimuthal drift speed,

$$v_\phi.$$

The code computes the theoretical value of

$$v_{\phi,\nabla B} + v_{\phi,c}$$

along the numerically computed gyrocenter trajectory and compares it to the measured gyrocenter precession from the full Boris orbit.

5. Extracting the numerical gyrocenter

The full particle trajectory contains rapid gyromotion plus slower guiding-center drift. To compare with drift theory, the code estimates the gyrocenter position using

$$\mathbf{R}_{gc} = \mathbf{x} + \frac{\mathbf{v} \times \mathbf{b}}{\Omega},$$

where

$$\Omega = \frac{qB}{m}.$$

The azimuthal position of the gyrocenter is then

$$\phi_{gc} = \tan^{-1}\left(\frac{Y_{gc}}{X_{gc}}\right).$$

The measured azimuthal drift speed is obtained from

$$v_{\phi,\mathrm{meas}} = R_{gc}\frac{d\phi_{gc}}{dt}.$$

Because numerical differentiation can still contain residual gyro-scale noise, the most robust comparison is not the instantaneous $v_\phi(t)$alone, but the integrated azimuthal precession,

$$\Delta\phi(t) = \int \frac{v_\phi}{R_{gc}}\,dt.$$

6. Results

The numerical run produced

$$B_{\mathrm{launch}} = 0.3273\ \mathrm{T},$$
$$\rho_L = 0.0640\ \mathrm{m}.$$

The mean measured azimuthal drift speed was

$$\langle v_{\phi,\mathrm{meas}}\rangle = -4.020 \times 10^3\ \mathrm{m/s}$$

while the mean theoretical drift speed was

$$\langle v_{\phi,\mathrm{theory}}\rangle = -3.986 \times 10^3\ \mathrm{m/s}$$

These mean values agree very well. The sign is also consistent: the gyrocenter precesses in the same azimuthal direction predicted by grad-B plus curvature drift theory.

The instantaneous drift-speed comparison remains somewhat noisy:

$$\mathrm{RMS}(v_{\mathrm{meas}} - v_{\mathrm{theory}}) = 6.682 \times 10^3\ \mathrm{m/s},$$
$$\mathrm{RMS}(v_{\mathrm{theory}}) = 4.381 \times 10^3\ \mathrm{m/s}$$

This does not mean the theory is failing. It mainly reflects the fact that the measured gyrocenter velocity is obtained by differentiating a finite-Larmor-radius orbit. Residual gyro-phase contamination remains in the instantaneous diagnostic.

The integrated precession comparison is much cleaner:

$$\Delta\phi_{\mathrm{meas}} = -0.4765\ \mathrm{rad},$$
$$\Delta\phi_{\mathrm{theory}} = -0.4713\ \mathrm{rad}.$$

The difference is only

$$\Delta\phi_{\mathrm{meas}} - \Delta\phi_{\mathrm{theory}} = -0.0053\ \mathrm{rad}.$$

Thus the cumulative azimuthal drift agrees very well with guiding-center theory.

In terms of the mirror point, for the 14 keV deuteron at 60°,

$$B_{\text{launch}} = 0.32730 \text{ T},$$

and the adiabatic mirror condition gives

$$B_{\text{mirror,theory}} = \frac{B_{\text{launch}}}{\sin^2 \alpha} = 0.43640 \text{ T}.$$

The corresponding theoretical mirror point is

$$z_{\text{theory}} = 1.12293 \text{ m}.$$

The numerical orbit gave:

$$z_{\text{gyrocenter}} = 1.10922 \text{ m},$$

$$z_{\text{particle}} = 1.11541 \text{ m}.$$

So the errors are:

$$\Delta z_{\text{gyrocenter}} = -1.37 \times 10^{-2} \text{ m},$$

$$\Delta z_{\text{particle}} = -7.52 \times 10^{-3} \text{ m}.$$

That is good agreement. The gyrocenter/particle distinction is expected because the particle's instantaneous z includes finite-Larmor-radius gyromotion, while the theory is really a guiding-center mirror condition.

7. Interpretation

The benchmark demonstrates the expected physics. The particle undergoes rapid gyromotion around the magnetic field while its gyrocenter slowly precesses azimuthally. This precession is produced by the combination of grad-B and curvature drift. The XY projection clearly shows the gyrocenter moving around the axis, while the XZ projection shows the particle bouncing in the mirror field.

The agreement between measured and theoretical integrated precession is the most important result. The instantaneous v_ϕ comparison is more sensitive to how the gyrocenter is extracted and how numerical derivatives are taken. The accumulated angle, however, averages over gyro-scale oscillations and provides a better test of drift physics.

The guiding-center approximation is approximately valid, but the drift is still large enough to see in the plots. The benchmark therefore strikes a useful balance: it is clean enough to compare to theory, but not so idealized that the drift becomes visually invisible.

8. Conclusion

This simplified analytical mirror provides a useful verification test for 3-D orbit codes. It checks more than simple gyromotion or mirror reflection. It tests whether the full particle pusher reproduces the slower guiding-center drift produced by magnetic-field gradients and curvature.

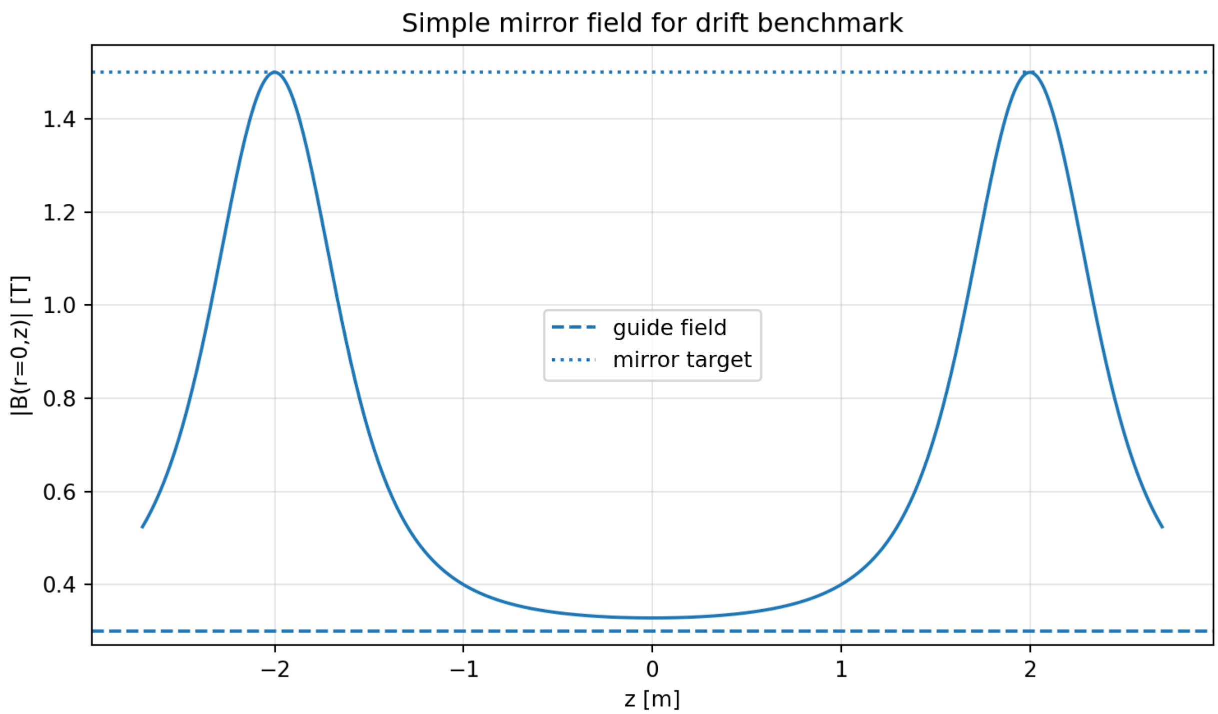
The key result is that the integrated azimuthal precession from the Boris orbit,

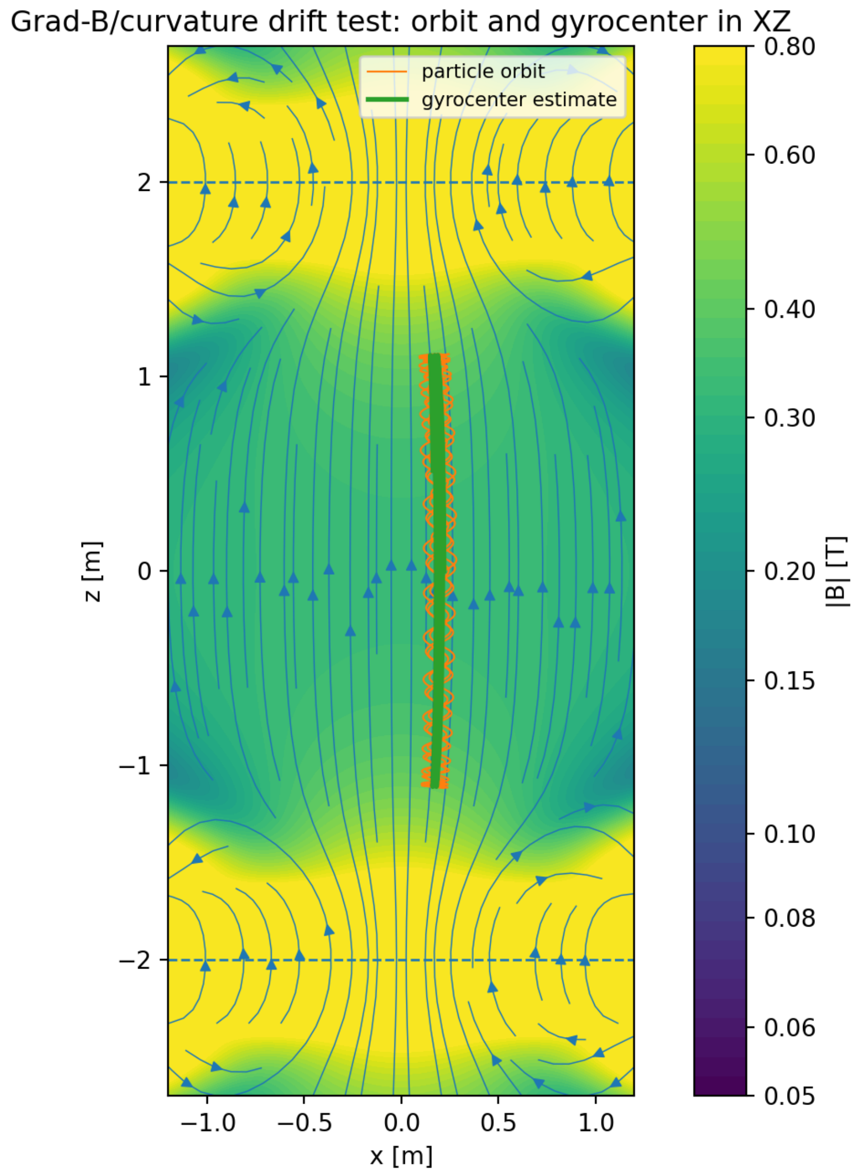
$$-0.4765 \text{ rad},$$

agrees closely with the grad-B plus curvature drift prediction,

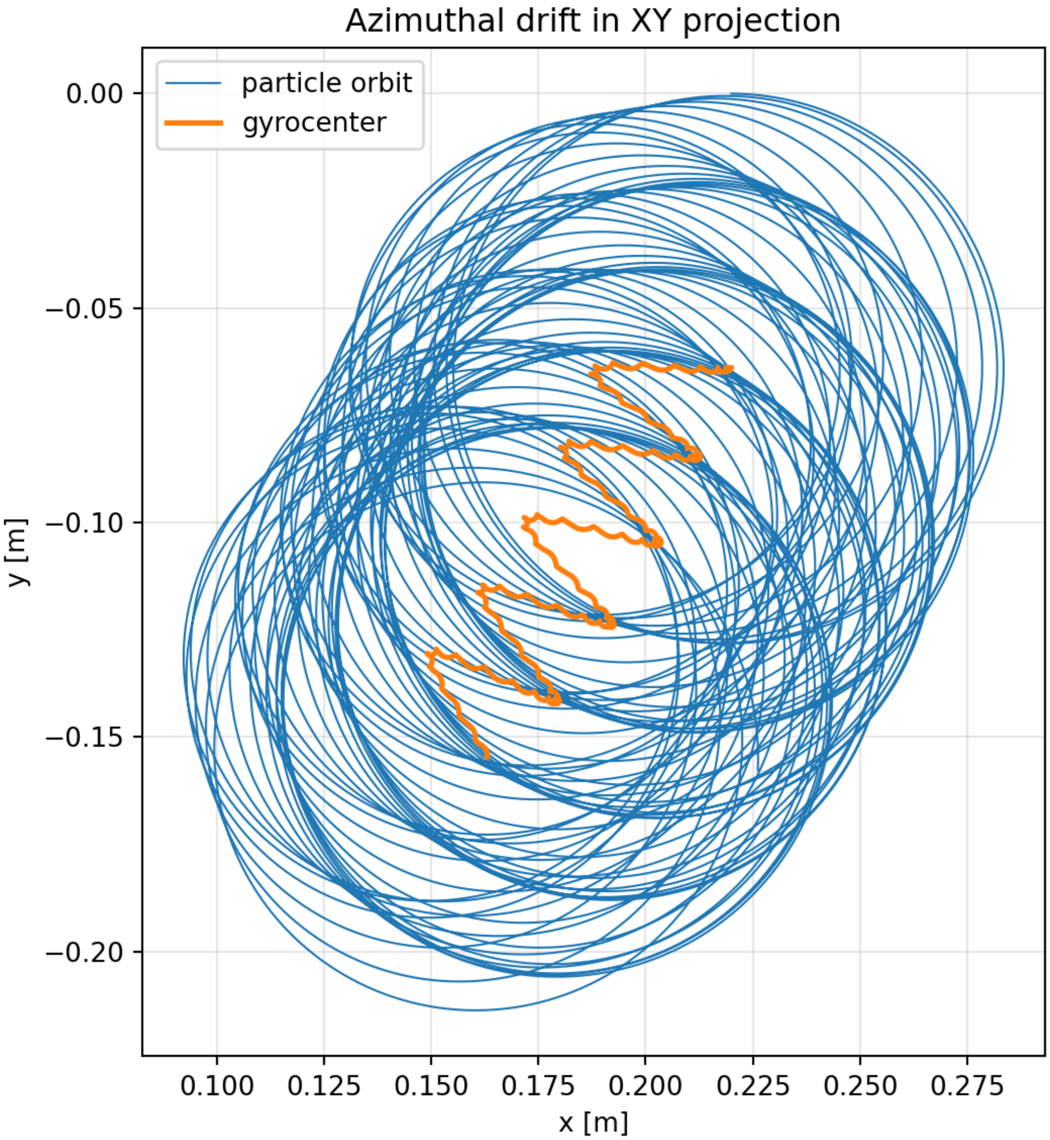
$$-0.4713 \text{ rad}.$$

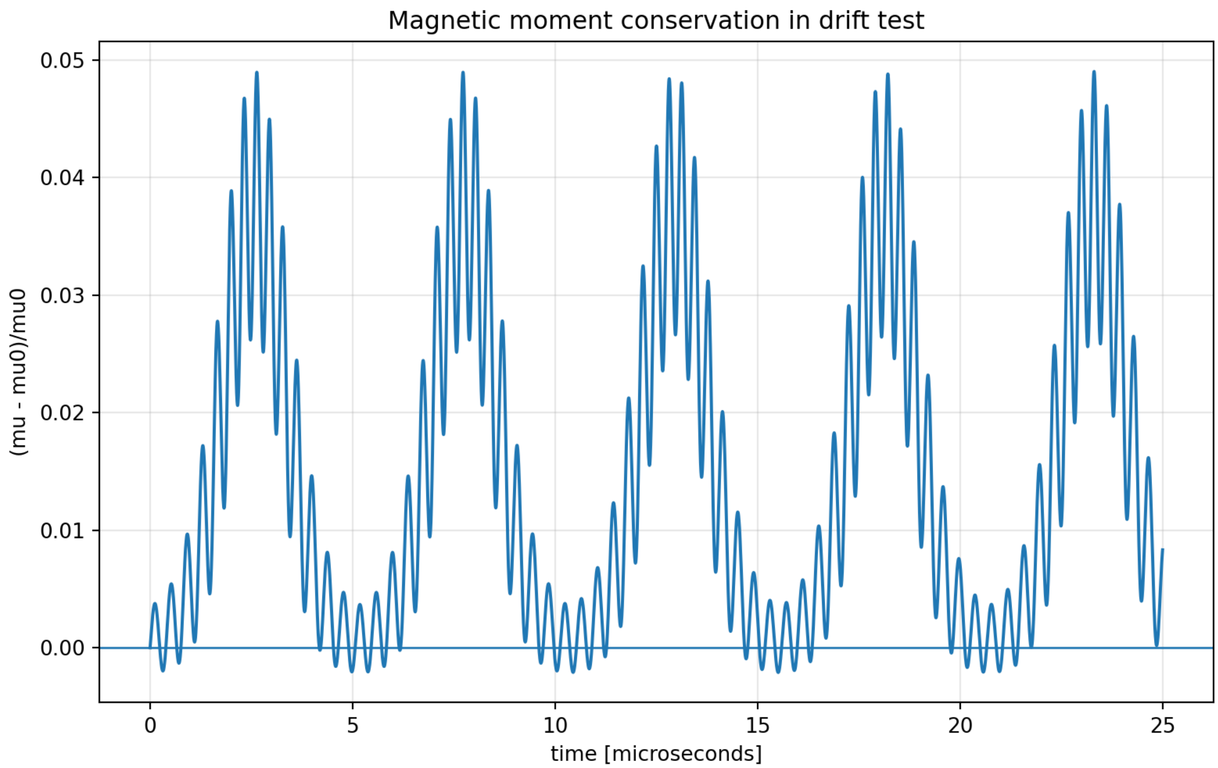
The remaining small difference is consistent with finite-Larmor-radius effects and residual gyrocenter-extraction noise. This makes the case a good intermediate benchmark between a purely uniform-field gyration test and a full realistic experimental mirror orbit calculation.

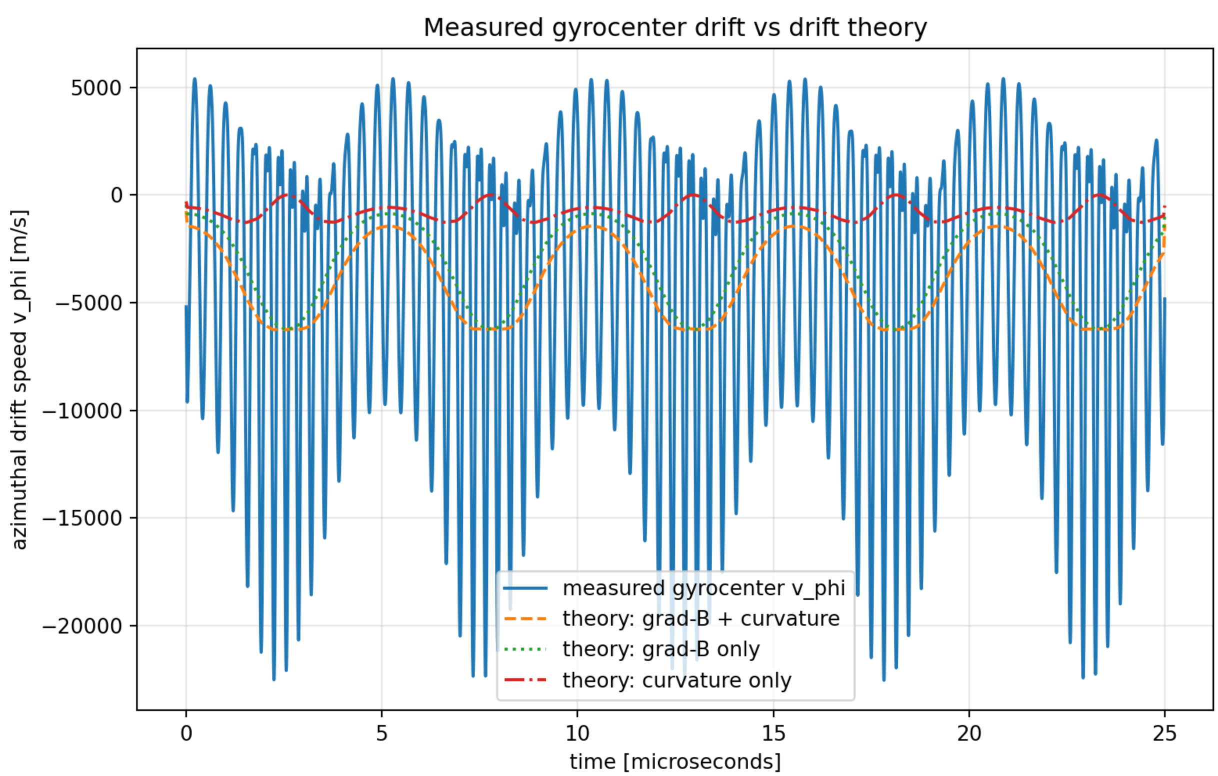


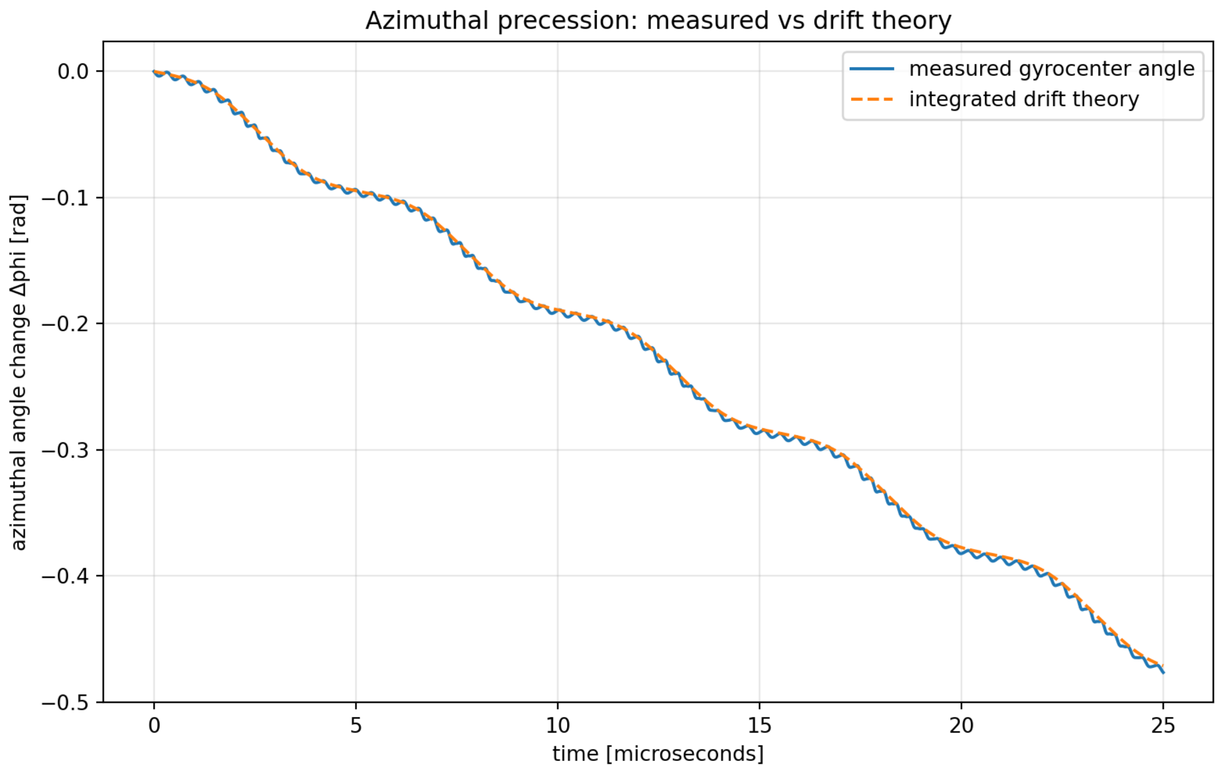


Grad-B/curvature drift test: orbit and gyrocenter in XZ
particle orbit
gyrocenter estimate
z [m]
x [m]
|B| [T]
2
1
0
−1
−2
−1.0
−0.5
0.0
0.5
1.0
0.80
0.60
0.40
0.30
0.20
0.15
0.10
0.08
0.06
0.05

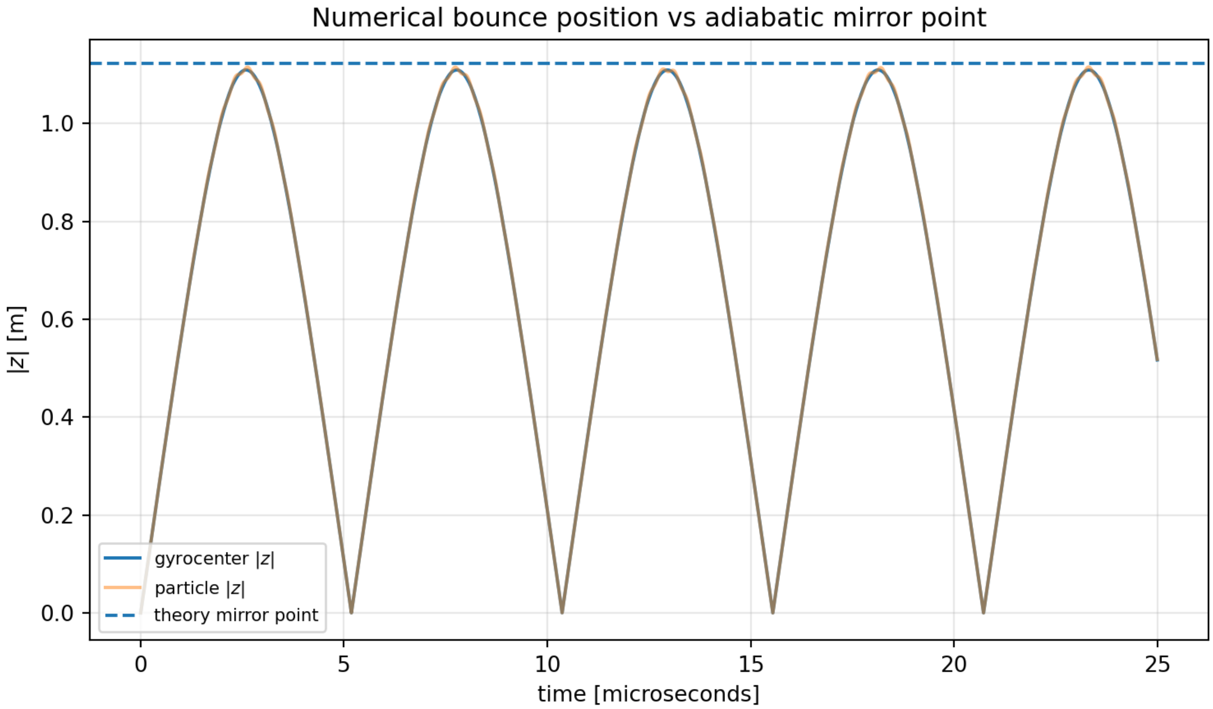

Azimuthal drift in XY projection
particle orbit
gyrocenter
0.00
−0.05
−0.10
−0.15
−0.20
y [m]
0.100
0.125
0.150
0.175
0.200
0.225
0.250
0.275
x [m]

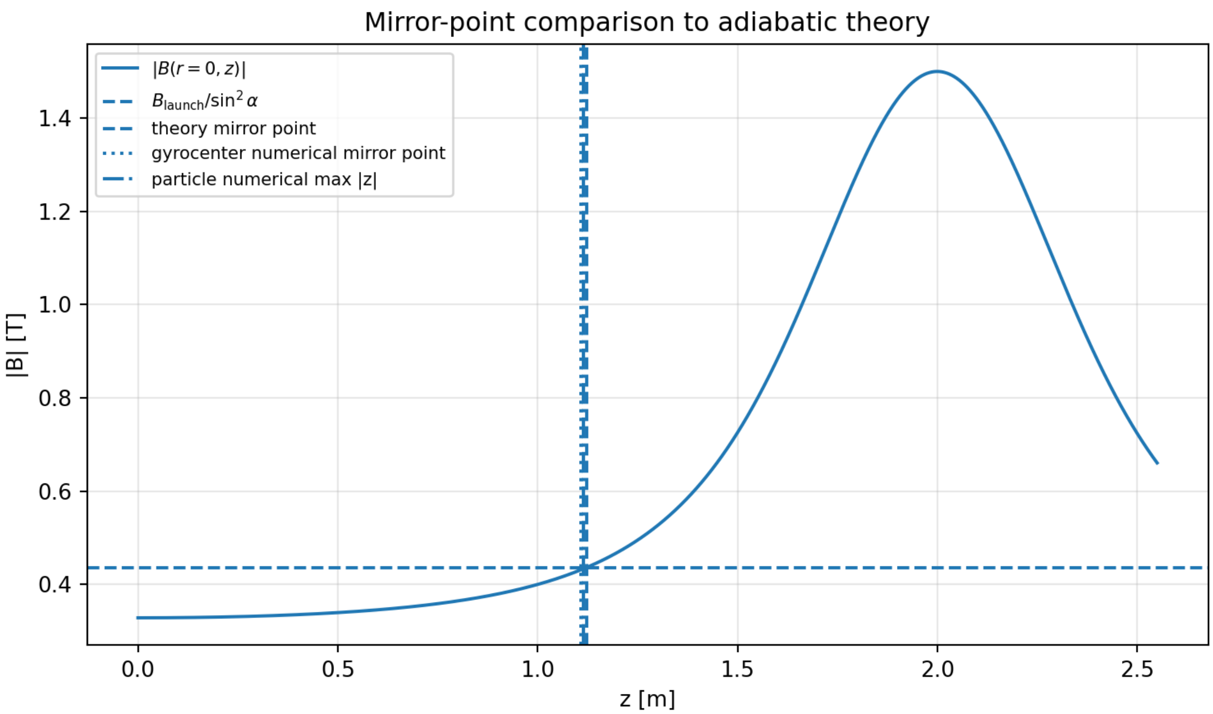

Magnetic moment conservation in drift test
(mu - mu0)/mu0
0.05
0.04
0.03
0.02
0.01
0.00
0
5
10
15
20
25
time [microseconds]


Measured gyrocenter drift vs drift theory
azimuthal drift speed v_phi [m/s]
5000
0
−5000
−10000
−15000
−20000
0
5
10
15
20
25
time [microseconds]
measured gyrocenter v_phi
theory: grad-B + curvature
theory: grad-B only
theory: curvature only



Azimuthal precession: measured vs drift theory
measured gyrocenter angle
integrated drift theory
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
azimuthal angle change Δphi [rad]
0
5
10
15
20
25
time [microseconds]


Numerical bounce position vs adiabatic mirror point
|z| [m]
time [microseconds]
gyrocenter |z|
particle |z|
theory mirror point
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
25


Mirror-point comparison to adiabatic theory
|B| [T]
z [m]
|B(r = 0, z)|
B_launch/sin² α
theory mirror point
gyrocenter numerical mirror point
particle numerical max |z|
0.4
0.6
0.8
1.0
1.2
1.4
0.0
0.5
1.0
1.5
2.0
2.5